

Real Analysis II

Tute-2

①

①

$$\sum_{n=0}^{\infty} \frac{x^n}{2^n}$$

$$a_n = \frac{1}{2^n} \quad \text{for all } n \in \mathbb{N} \quad (x=0)$$

$$\lambda = \limsup |a_n|^{1/n} = \limsup \left| \frac{1}{2^n} \right|^{1/n}$$

$$= \limsup \left| \frac{1^{1/n}}{2^{n^{1/n}}} \right|$$

$$= \limsup \left| \frac{1}{2} \right|$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} = (0, \infty)$$

$$\therefore 0 < \lambda < \infty$$

$$R = \frac{1}{\lambda} = 2$$

Interval of Convergence is

$$IC = [-R, R] \\ = [-2, 2]$$

$$iv) \sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot \dots (2n-1)}{(n+1)!} x^{2n}$$

$$; IC = 0$$

$$a_n = \frac{1 \cdot 3 \cdot \dots (2n-1)}{(n+1)!}$$

$$a_{n+1} = \frac{1 \cdot 3 \cdot \dots (2n+1)}{(n+2)!}$$

$$1/R = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

$$= \sqrt{\lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdot \dots (2n+1)}{(n+2)!} \times \frac{(n+1)!}{1 \cdot 3 \cdot \dots (2n-1)}}$$

$$= \sqrt{\lim_{n \rightarrow \infty} (2n+1)}$$

$$= \infty$$

~~2021/1/1~~

$$\frac{1}{R} = \infty$$

$$R = \frac{1}{\infty}$$

$$= 0$$

$$I_C = [C-R, C+R]$$

$$= [0, 0]$$

$$2. \sum_{n=0}^{\infty} \frac{(n+1)}{(n+2)(n+3)} x^n \quad ; \quad C=0$$

$$R/z \quad a_n = \frac{(n+1)}{(n+2)(n+3)}$$

$$a_{n+1} = \frac{(n+2)}{(n+3)(n+4)}$$

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(n+1)}{(n+2)(n+3)} \cdot \frac{(n+3)(n+4)}{(n+2)} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{(1+1/n)(1+4/n)}{(1+2/n)(1+3/n)} \right|$$

$$= \lim_{n \rightarrow \infty} 1$$

$$I_C = [C-R, C+R]$$

$$= [-1, 1] //$$

$$(3) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n \quad ; \quad c=0$$

$$a_n = \frac{(-1)^{n+1}}{n}$$

$$a_{n+1} = \frac{(-1)^{n+2}}{n+1}$$

$$R = \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1}}{n} \times \frac{n+1}{(-1)^{n+2}} \right|$$

$$= \lim_{n \rightarrow \infty} \frac{n+1}{n}$$

$$= \lim_{n \rightarrow \infty} 1 + \frac{1}{n}$$

$$= 1$$

$$I.C = [c-R, c+R]$$

$$= [-1, 1]$$

$$(4) (i) \sum_{n=1}^{\infty} 3x^{3n-1}$$

$$\sum_{n=1}^{\infty} 3(x^3)^{n-1}$$

$$S = \frac{a}{(1-r)}$$

$$= \frac{3}{1-x^3}$$

; closed form of the power series

$$a = 3 \quad r = x^3$$

$$\sum_{n=1}^{\infty} 3x^{3n-1} = \frac{3}{1-x^3}$$

The formula is valid for all $|x^3| < 1$, which means that absolute value of x^3 must be less than 1
 $\therefore$ open interval $(-1, 1)$; largest set

$$ii) S(x) = \sum_{n=0}^{\infty} \frac{x^{3n}}{n+1}$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

In here $a = -x^3$ so;

$$S(x) = \sum_{n=0}^{\infty} \frac{(-x^3)^n}{n+1} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n}}{n+1}$$

Now write in closed form

$$S(x) = - \sum_{n=0}^{\infty} \frac{(-x^3)^n}{n+1} = - \sum_{n=0}^{\infty} (-1)^n \frac{x^{3n}}{n+1}$$

The closed form of this power series is

$$S(x) = -\ln(1+x^3)$$

for valid;

$$1+x^3 > 0$$

$$x^3 > -1 //$$

$$C_n = \frac{f^n(0)}{n!}$$

⑥ $f(x) = \ln(1+x)$

$$n=0; f(x) = \ln(1+x), f(0)=0 \Rightarrow C_0=0$$

$$n=1; f'(x) = \frac{1}{1+x}, f'(0)=1 \Rightarrow C_1=1$$

$$n=2; f''(x) = -\frac{1}{(1+x)^2}, f''(0)=-1 \Rightarrow C_2 = -\frac{1}{2}$$

$$n=3; f'''(x) = \frac{2}{(1+x)^3}, f'''(0)=2 \Rightarrow C_3 = \frac{2}{3!} = \frac{1}{3}$$

$$n=4; f^{(4)}(x) = -\frac{6}{(1+x)^4}, f^{(4)}(0)=-6 \Rightarrow C_4 = \frac{-6}{4!} = -\frac{1}{4}$$

Thus $\ln(1+x) = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + \dots$

$$= 0 + 1x + (-)\frac{1}{2}x^2 + \frac{1}{3}x^3 + (-)\frac{1}{4}x^4 + \dots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

where $|x| \leq 1 \Rightarrow -1 \leq x \leq 1$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$x=1$

$$\ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$